Macroeconomic Trends and Business Data Combined: Inflation, Demand, and Central Bank Actions Mapped

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Abstract

One way economists struggle is when they look at big picture numbers too separately. That gap makes it harder to see links between price changes, what people buy, and central bank moves across months or years. Instead of siloing data sources, this work builds a connected system pulling federal statistics into one framework. A live updating Power BI interface brings clarity and is enhanced using Deneb to shape complex patterns visually. Time unfolds from 2016 into early 2026, tracking five linked areas: breakdowns within inflation itself, shifts in household outlays, movement in after tax earnings, job market signals mixed with public outlook, and delayed reactions in interest rate decisions.

From official public sources, data were gathered and then arranged by month. Following that, changes compared to the same months in previous years were calculated. To allow comparison across series, all indices were rescaled using 2016 as a reference point. A measure of inflation delayed by three months was built to examine how policy adjusts over time. Transportation costs stood out due to sharp swings in the near term. In contrast, housing related prices changed more slowly but lasted longer in their impact. Spending patterns tracked closely with personal income when viewed over many years. Brief mismatches appeared during crisis periods yet faded afterward. Unemployment returned to typical levels faster than people's overall outlook on the economy. This gap hints that lasting price pressures affected confidence beyond job availability. Near the end, a detailed visual chart featuring a trend line revealed that higher interest rates followed earlier spikes in inflation. That pattern implies monetary policy decisions relied more on past inflation readings than on current ones.

What emerges is a clear sense of how business intelligence tools help shift focus from basic visuals to more connected explanations. Instead of stopping at charts, the work uses dashboards, calculated indicators, and an original graphic form not explored fully during lessons. Through these pieces, insight grows into how economic patterns unfold over time. The approach reveals pathways within BI processes that encourage stronger interpretation of complex shifts.

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Conference'17, Washington, DC, USA
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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Keywords

business intelligence, macroeconomics, inflation, consumer spending, disposable income, monetary policy, Power BI, Deneb, data visualization

ACM Reference Format:
Silas Curry. 2026. Macroeconomic Trends and Business Data Combined: Inflation, Demand, and Central Bank Actions Mapped. In . ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Most times, turning scattered information into useful insight defines how organizations make choices. Macroeconomic figures fit neatly here due to their time based patterns, linked factors, and practical consequences. While inflation hits some industries harder than others, shifts in buying habits rarely follow prices step by step. Because delays happen more often than immediate reactions in central bank moves, timing matters greatly. When processes merge model creation with calculated metrics, visual tools become helpful alongside explanation layers. Such setups handle complexity without losing clarity across stages.

This study explores a specific research question. Focusing on this inquiry guides the analysis. The aim is understanding through careful examination. Answers emerge by reviewing evidence systematically.

> Over time, inflation shapes how much people spend, while their leftover earnings shift with interest rate moves. Yet changes in spending feed back into price trends, creating a loop. Instead of viewing these pieces separately, linking standard reports with tailored visuals reveals delayed effects from past central bank actions. When policy adjusts, its impact trails, sometimes by months, before showing up in buying habits. Looking beyond snapshots uncovers patterns dashboards alone might miss. Timing matters just as much as direction when tracking economic cause and effect.

One look at the layout shows a Power BI setup spread over several pages, built using calculated fields along with a single Deneb chart for deeper insight. Though focused on multiple outcomes, it highlights how lately, consumer outlays have risen faster than both take home pay and overall price increases. Transportation costs stood out due to wider swings compared to other parts of the CPI basket. Over extended periods, trends in earnings and expenditures move closely together. In the custom built Deneb plot, past inflation levels link up clearly with where interest rates sit today. Each of these observations appears directly within the different sections of the provided report visuals. [2]

What comes next is outlined here. Section 2 covers where the data came from along with how it was combined. Section 3 walks through the approach used for analysis and the measures created. Section 4 displays findings for each dashboard screen while making sense of key trends. Section 5 examines the more complex Deneb visualization and explains how it helps analysis. Section 7 discusses project constraints. Section 8 concludes the paper.

2 Data Sources and Integration

2.1 Source Selection

Monthly U.S. macroeconomic data come from official public repositories for this study. From the Bureau of Labor Statistics Consumer Price Index program, inflation figures include both overall CPI U and specific categories such as food, shelter, and transportation. CPI is a calculation tracking how prices shift over time for typical urban household purchases. Sources supporting these definitions appear in the cited references.

From the U.S. Bureau of Economic Analysis came figures on consumer spending and disposable income. The agency counts what people buy directly or have bought for them, labeling it personal consumption expenditures. Updates appear regularly in its Personal Income and Outlays report, where disposable personal income also appears. These numbers form a core part of how household economic activity is measured. Sources include official BEA publications tied to spending and income tracking.

From FRED, data on monetary policy came through the Federal Funds Effective Rate, while labor market figures used the civilian unemployment rate. The federal funds rate appears in FRED as the overnight interest level at which banks exchange reserves held at Federal Reserve Banks. Sources include [1, 11].

Measured through ongoing surveys, consumer sentiment appears in the University of Michigan index, capturing how households view their financial outlook. This indicator draws from regular responses about economic confidence and personal spending conditions.

These sources were selected for several reasons. First, they are publicly documented and widely recognized. Second, they are published monthly with similar regularity. Third, they reflect linked aspects of the economy, including inflation, consumer mood, spending patterns, income, policy actions, and labor market conditions. Finally, their reliable structure and accessibility make them well suited for integrated business intelligence models.

2.2 Temporal Scope and Unit of Analysis

Beginning in January 2016 and running into early 2026, the analysis captures phases such as pre pandemic stability, disruption during the health crisis, rising prices, tighter central bank policy, and more recent easing trends. A monthly frequency remains in place since it keeps fine grained movements visible. This helps make sense of how quickly inflation shifted, consumption patterns rebalanced, and policy effects unfolded over time.

2.3 Data Preparation

Data preparation involved several stages:

(1) Monthly data were cleaned and aligned using a shared date format.
(2) Type correction involved converting date entries into proper date values and quantitative measures into numeric form. Fields that were previously mismatched were recast into their intended data types. This conversion fixed inconsistencies across time related columns and ensured that numeric values were interpreted correctly.
(3) Year over year changes were calculated for CPI, consumer outlays, and household earnings. These changes were computed by comparing each value to the same month in the prior year. Growth patterns became visible once data points aligned across twelve month intervals. The results were expressed as percentages, showing expansion or contraction clearly.
(4) Starting from a base where 2016 equals 100, major indexes were rescaled to match one baseline. Because of this shift, tracking growth differences became more straightforward across datasets.
(5) Lag construction began by shifting inflation data backward by three months. This adjustment allowed observation of later policy shifts in the visuals. Using past values helped reflect how decisions respond slowly and matched the timing between cause and effect more closely.

The unified table became the foundation for Power BI calculations, key performance indicators on report pages, and the specialized Deneb visualization.

3 Methodology

3.1 Analytical Design

Starting with patterns already present, the study applied descriptive methods to map out scale, expansion, and movement within each data series. After that, deeper investigation unfolded by contrasting elements, breaking down components, adjusting values, and examining delayed effects. This second phase moved past obvious shifts using structured comparisons rather than simple summaries.

The workflow was organized into five dashboard pages:

(1) Executive overview
(2) Inflation decomposition
(3) Income versus spending
(4) Economic context
(5) Advanced Deneb analysis

This layout was intentional. Instead of packing every metric into a single view, each screen addressed a specific analytical task while still linking logically to the others. A clear thread ran through them all.

3.2 Derived Metrics

Several metrics built from raw data were essential to the analysis.

Year over year growth rates. One reason yearly comparisons were used in place of monthly ones within key visuals is that they smooth out brief fluctuations. These adjustments help highlight broader patterns over time. Clarity drove this approach by making it easier to identify shifts that matter without distraction.

$$YoY_t = \frac{X_t - X_{t-12}}{X_{t-12}} \times 100$$

Here, X_t represents the value of a series during month t. Each point ties directly to its monthly position in time.

Index normalization. Starting from 2016, CPI, spending, and disposable income were adjusted to a shared reference point. Though measured in different units, they were placed on equal footing. This made it possible to compare relative movement across distinct scales and removed the distortion caused by uneven starting levels.

$$Index_t = \frac{X_t}{X_{2016\ baseline}} \times 100$$

Because of this transformation, comparing total growth over time became straightforward across variables.

Lagged inflation. Three months prior to each policy moment, inflation data shaped the context for that month's federal funds rate. Though timing shifts may seem minor, comparing today's interest decisions with older price trends reveals delayed reactions. Instead of aligning rates with current pressures, earlier readings provided a better frame for interpretation. This adjustment exposes how past economic signals influence present settings.

3.3 Visualization Strategy

A mix of familiar reporting elements appears alongside a single specialized component on the dashboard. These include trend graphs tracking values over time, scaled comparison plots, summary indicators showing key metrics, and commentary enhanced information panels. Built using Deneb, a verified add on for Power BI that relies on Vega Lite syntax, the advanced visual takes shape. References support both the broader customization framework and the specific tool's documentation [3, 4].

Because the assignment called for a sophisticated visual beyond what we studied in detail, Deneb was an appropriate choice. More importantly, it had analytical value. Examining whether elevated past inflation aligns consistently with increased interest rates works better with a scatter plot and trend line than with a standard time series display.

4 Results

4.1 Executive Overview

The top level overview shows three real time KPI snapshots along with a pair of comparison growth displays. Latest figures from the final dashboard indicate consumer spending rose by 5.34%, while disposable income climbed at 3.90%. Inflation, meanwhile, stands at 2.43%. [2]

Figure A in the Appendix shows the executive overview dashboard showing key performance indicators including year over year inflation, consumer spending, and disposable income growth, along with comparative trend visualizations.

The initial chart pairs inflation against consumer spending trends, while the second sets disposable income beside spending changes. Even as prices rise more slowly, spending continues climbing faster than both income and inflation, which signals steady consumer activity. This pattern appears in the dashboard summary and highlights how demand remains strong despite cooler price increases [2].

This point matters in analysis for two reasons. First, overall demand stayed steady lately even while inflation slowed down. Second, price levels on their own do not fully explain what people buy. Recent purchases may reflect rising incomes, remaining savings buffers, or lingering shifts in habits after earlier economic disruptions.

4.2 Inflation Decomposition

The inflation page splits CPI into four parts: headline CPI, food, shelter, and transportation. At its highest point, overall prices rose by nearly 9 percent. Food climbed even faster, reaching just over 11 percent at the peak. Shelter followed behind, rising above 8 percent during that period. Meanwhile, transportation surged past the others, topping out near 23 percent. These figures come from the dashboard output [2].

Figure A in the Appendix illustrates overall CPI alongside food, shelter, and transportation components, highlighting differences in volatility and persistence across categories.

Inflation clearly broke into distinct patterns across sectors. Transportation swung the most, with sudden jumps followed by quick drops, matching behavior commonly seen in energy linked areas. In contrast, shelter stayed elevated longer instead of spiking sharply, which reflects how slowly housing costs adjust. Food prices climbed significantly, but their movements were still less extreme than those in transportation.

This difference between volatility and persistence matters. A sudden spike in headline inflation may come from unstable parts of the index, yet the policy implications change when slower moving sectors such as shelter begin to rise. The dashboard correctly shows that transportation swung hardest, while shelter changed more slowly and more persistently [2].

4.3 Income and Consumer Spending

The income versus spending page explores patterns across different time frames. The upper chart compares annual changes in disposable income and consumer spending. The lower chart compares both measures from a shared baseline set in 2016. Though these two trends usually follow similar paths, breaks emerge when economic conditions shift unexpectedly [2].

Figure A in the Appendix shows both year over year growth rates and indexed comparisons normalized to a 2016 baseline.

The indexed panel reveals especially clear patterns. Throughout the full period, growth in disposable income and consumer spending outpaced inflation measured by CPI, suggesting that household resources and expenditures increased faster than the overall price level. What stands out most is how sharply conditions shifted during the pandemic. Government transfers and related policies pushed income upward almost immediately, even though people cut back on purchases at first. Spending only picked up later and rose quickly once movement became easier again. This reflects how unusual conditions disrupted normal behavior. Extra savings built during restrictions were eventually released into purchases. As life returned closer to pre pandemic routines, the link between earnings and spending reappeared.

The resilience of aggregate demand is supported not just by the latest values, but also by the combined evidence of long term index alignment and temporary short term divergence [2].

4.4 Economic Context Variables

Inflation trends appear alongside jobless figures and public sentiment indicators so the broader economic picture can be seen more clearly. As prices climbed, authorities responded with tighter monetary conditions, and rate hikes unfolded quickly once inflation pressures became clear. Even when employment stopped deteriorating, people felt less confident about finances, revealing a gap between labor data and consumer outlooks. Evidence supporting this sequence appears in the dashboard output [2].

Figure A in the Appendix combines inflation trends, federal funds rate, unemployment, and consumer sentiment to provide macroeconomic context.

Following the pandemic, unemployment spiked sharply before slowly returning toward normal. After staying near zero for some time, interest rates climbed quickly once inflation began rising faster. Consumer sentiment worsened noticeably during the inflation surge and has stayed below earlier levels since then.

This combination supports a meaningful economic interpretation. Even as job markets steadied, confidence stayed low because rising prices reduced what households could afford and changed how they viewed future conditions. Put differently, consumer sentiment reacted more sharply to inflation than to unemployment during the later part of the period studied.

5 Advanced Visualization and Lag Analysis

The advanced page uses Deneb to build a scatter plot showing the federal funds rate against inflation delayed by three months, along with a regression line. On the final dashboard screen, the upward trend of the fitted line shows a link between elevated past inflation and increased interest rates. Groupings of data points suggest not a smooth adjustment path, but separate phases in how monetary policy operated. Evidence supporting this interpretation appears in [2].

Figure A in the Appendix shows the relationship between three month lagged inflation and the federal funds rate with a fitted regression line, revealing delayed policy response patterns.

This visual contributes more than novelty. Standard line charts show how inflation and rates move across time, but they do not directly test whether interest rates align better with earlier inflation levels than with current ones. Moving into scatter plot space allows a clearer look at the relationship, and fitting a trend line helps evaluate its direction and general strength.

The upward tilt of the line suggests policy delay. Although this does not prove causality in a strict statistical sense, it supports the view that interest rates tracked past inflation more closely than same month inflation. The visible clusters reinforce that point. Low rate clusters appear when earlier inflation was relatively mild, while higher rate clusters emerge during periods of elevated lagged inflation and tightening.

One reason this Deneb visual stands out is that it required a separate Vega Lite specification. Built around lag analysis, it does more than decorate the page. It performs an analytical function that was not explored deeply in class and aligns closely with the expectation for a more advanced custom visual [3, 4].

6 Discussion

Overall, the dashboard points to four wider takeaways.

First, inflation should not be treated as one uniform trend. Short term movement often swings sharply because of transportation costs, especially over brief periods. In contrast, shelter contributes more persistent pressure and drags its influence across months or years. This split matters because short lived shocks call for different interpretation than deeply rooted increases.

Second, consumer spending tends to follow disposable income fairly closely over time. Although the pandemic briefly pulled them apart, growth tracking reveals a pattern that returns toward alignment. The disruption was clear, but the evidence does not support lasting separation between earnings and expenditure.

Third, despite falling unemployment, rising prices took a toll on consumer confidence. When inflation climbed, people felt worse off. This suggests that economic well being depends not just on employment, but also on whether household income retains purchasing power.

Fourth, lag analysis sharpens interpretation of the monetary tightening phase. Rather than tracking current inflation alone, the federal funds rate responded more clearly to past inflation levels. This is visible in the Deneb regression chart. Time shifting the data reveals structure that ordinary summaries can miss.

7 Limitations

This project has several limitations.

- The findings are observational and exploratory. They point to patterns without establishing causality.
- The three month lag served as a practical analytical window and was not selected through formal optimization.
- Fiscal transfers, oil prices, and job sector composition were not included in the model.
- The fitted line in the Deneb visual summarizes trends descriptively and is not intended as a full econometric model.

Even so, the project's core value remains intact despite these limits. Its aim was to demonstrate strong business intelligence practice through integrated modeling, calculated measures, interactive storytelling, and one advanced analytical visual. On those terms, the workflow achieves what it set out to do.

8 Conclusion

This project shows the use of business intelligence software not just for making separate graphs, but for linking broader economic trends together. Instead of presenting data in pieces, it pulls several government sources into one cohesive framework. Through calculated metrics such as growth rates and indexed values, patterns begin to emerge clearly. A multi section Power BI interface organizes these findings logically across views. With a Deneb based regression visual added, deeper relationships within the data come into focus. Explanation replaces mere description.

Although inflation varied across categories throughout the observed timeframe, consumer expenditures followed disposable income steadily over time. Even as job markets improved, public

confidence dipped noticeably. Policy adjustments aligned more closely with past inflation data than with current readings. When metrics are built thoughtfully and visuals are used carefully, analytics platforms can support deeper insight into economic patterns. Such outcomes emerge not from technology alone, but from how numbers are framed alongside context.

Looking beyond the specific results, this work reflects what Applied Business Intelligence aims to do. It moves past displaying numbers and instead weaves them together, reshapes their meaning, makes sense of patterns, and communicates insights clearly so that decisions gain context and interpretation becomes stronger.

A Appendix

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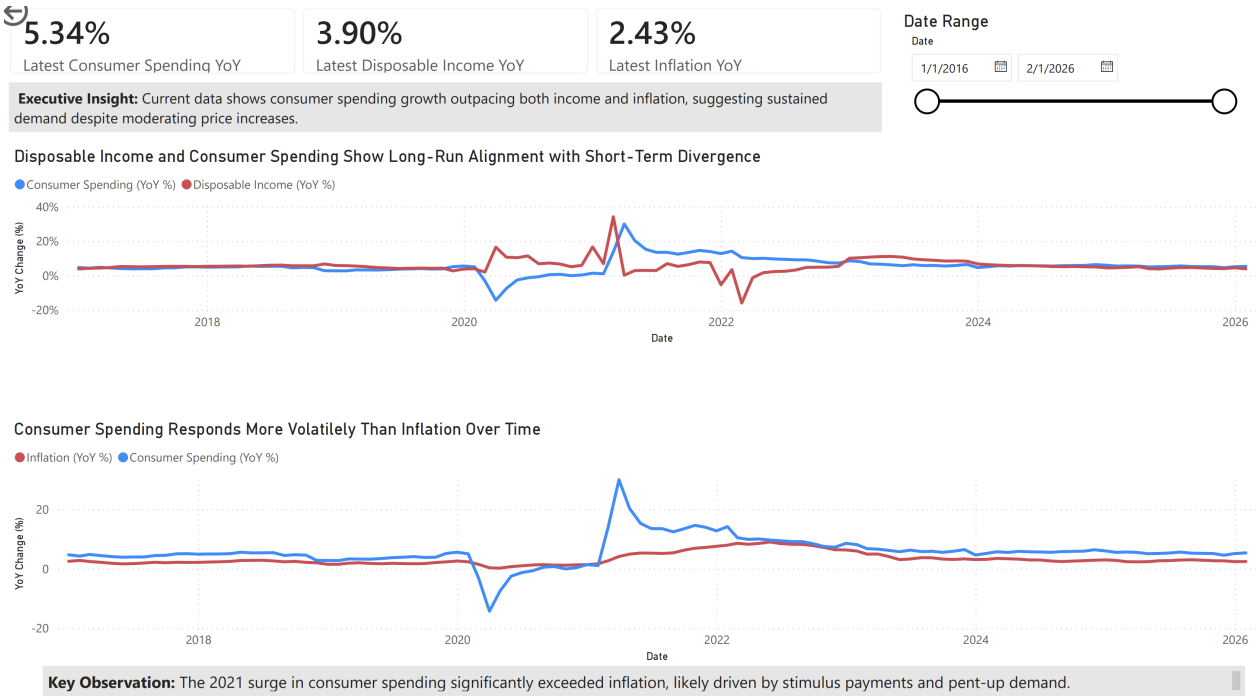


Figure 1: Executive overview dashboard showing key performance indicators including year over year inflation, consumer spending, and disposable income growth, along with comparative trend visualizations.

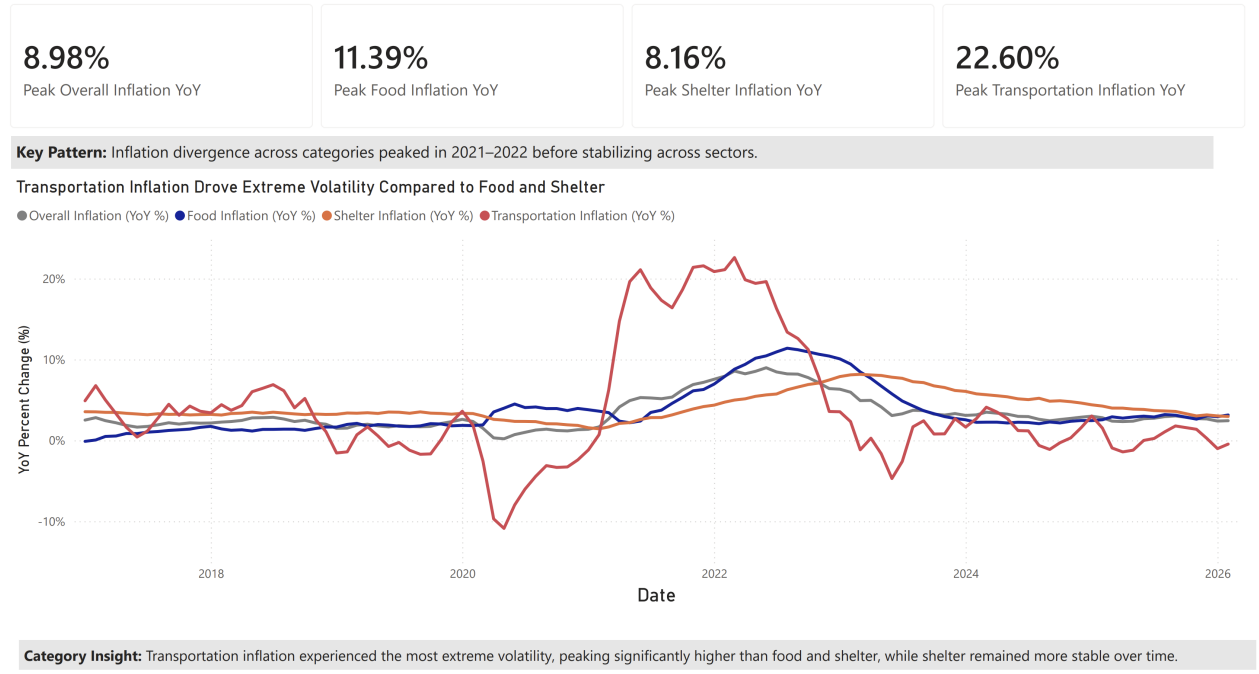


Figure 2: Inflation decomposition dashboard illustrating overall CPI alongside food, shelter, and transportation components, highlighting differences in volatility and persistence across categories.

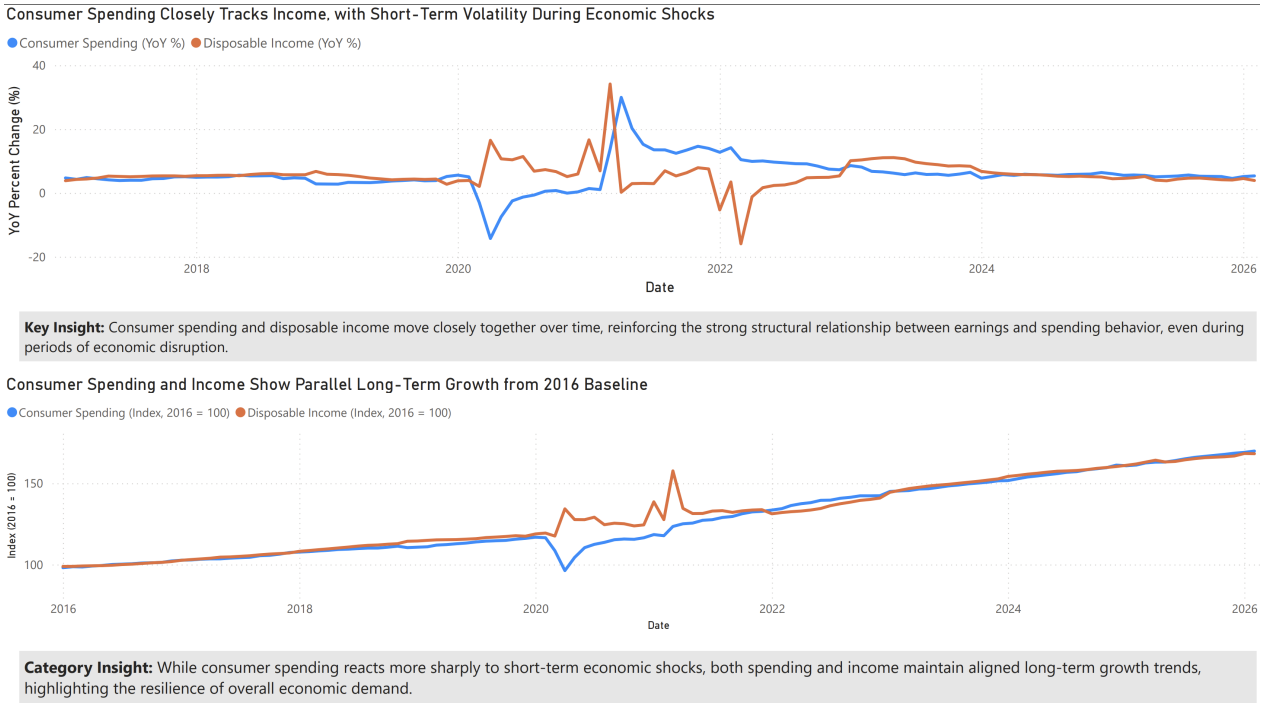


Figure 3: Income versus spending dashboard showing both year over year growth rates and indexed comparisons normalized to a 2016 baseline.

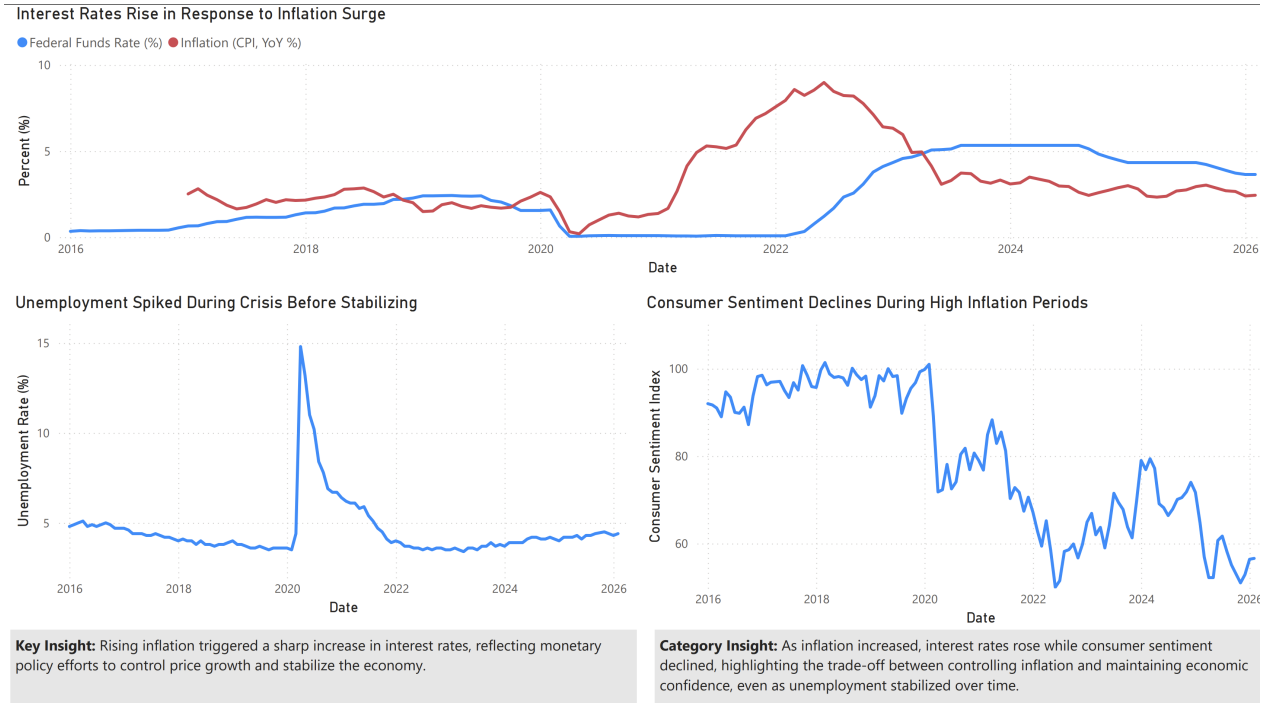
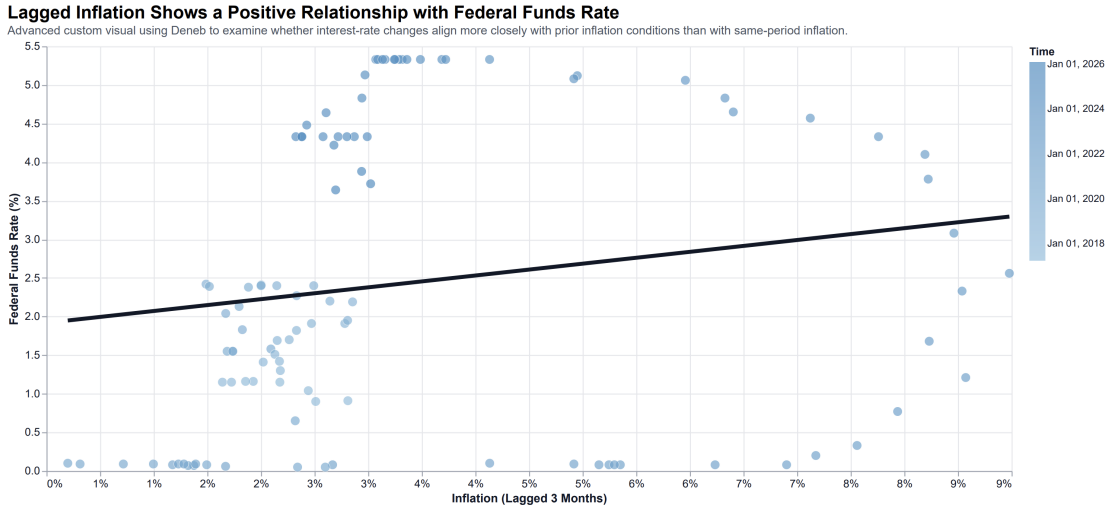


Figure 4: Economic context dashboard combining inflation trends, federal funds rate, unemployment, and consumer sentiment to provide macroeconomic context.



Advanced Insight: The positive slope of the regression line indicates that higher lagged inflation is generally associated with higher federal funds rates. This supports the interpretation that monetary policy responds to prior inflation conditions with a delay, rather than adjusting immediately. The clustering of points further suggests that policy responses occur in distinct economic regimes rather than as a continuous linear adjustment.

Additional Insight: Periods of near-zero interest rates appear even at moderate inflation levels, reflecting policy constraints during expansionary phases, while higher-rate clusters correspond to tightening cycles following inflation surges.

Figure 5: Deneb custom visualization showing the relationship between three month lagged inflation and the federal funds rate with a fitted regression line, revealing delayed policy response patterns.